

Kasra Ahmadi

Machine Learning | LLM Systems | ML Security & Privacy | Post-Quantum Cryptographic Hardware

kasra.research@gmail.com | [Google Scholar](https://scholar.google.com/citations?user=KasraAhmadi) | github.com/KasraAhmadi | [linkedin.com/in/kasra-ahmadi](https://www.linkedin.com/in/kasra-ahmadi) | Sunnyvale, CA

✓ **Authorized to work in the U.S. - No employer sponsorship required**

PROFESSIONAL SUMMARY

Ph.D. candidate in Computer Science and Engineering with **10+ peer-reviewed papers**, working in machine learning and security. On the ML side, trains and deploys language-model systems for privacy and safety - most recently a **PII (personal data) detection model** that outperformed OpenAI's Privacy Filter, and the serving system behind it. Earlier research built **fault-detection and side-channel defense schemes** that protect cryptographic hardware and software - including post-quantum standards - from physical attacks, reaching near-complete fault coverage at about 1% overhead on ARM chips and FPGAs. Best Paper Award, IEEE S&P Workshops 2025.

EDUCATION

Ph.D. in Computer Science and Engineering *University of South Florida, Tampa, FL* Jan 2022 - Oct 2026 (expected)
Machine Learning, Privacy-Preserving ML, Side-Channel & Fault Analysis, Post-Quantum Cryptography | **Best Paper Award, IEEE S&P Workshops 2025**

Master of Information Technology Engineering *Amirkabir University of Technology, Tehran, Iran* Sep 2018 - Aug 2021

Bachelor of Science in Computer Engineering *Isfahan University of Technology, Isfahan, Iran* Sep 2012 - Jul 2017

EXPERIENCE

RealmLabs May 2026-Aug 2026
ML Research Engineer Intern *Sunnyvale, CA*

- Built the PII (personal data) detection model behind [RealmGuard](#): a 634K-parameter head over a frozen Qwen3.5-9B that finds and labels sensitive information in text. A **multi-head attention** architecture reads the hidden states of the whole text at once - outside the causal model - removing the causality limitation that stops a decoder LLM from using future context. Beat OpenAI's Privacy Filter on a shared benchmark, **0.837 vs. 0.654 mean micro-F1**.
- Consolidated text generation, PII detection, sparse autoencoders, and multiple classifier **probes** into one service over a single GPU-resident model; using a customized vLLM engine on an A100, a **single forward pass** returns the generation output together with every probe at **~200 ms latency under 10 concurrent requests**.
- Scaled generation throughput from **64.4 to 267.0 tokens/s** on the same A100 through continuous batching and prefix caching.

University of South Florida Jan 2022-Present
Ph.D. Researcher (NSF-funded) *Tampa, FL*

- Machine learning & privacy:** Won a **Best Paper Award** (IEEE S&P Workshops 2025) for a method that fine-tunes language models on resource-constrained IoT devices with minimal available RAM, without sharing their private data, using differential privacy and federated learning.
- Built MAED, a method that catches fault-injection attacks on neural networks while they run, by checking math properties of their activation functions.
- Studied adversarial attacks that quietly lower the accuracy of RAG (retrieval-augmented) AI systems.
- Cryptography & hardware security:** Designed fault-detection and side-channel defense schemes for classical and post-quantum cryptography (RSA, elliptic-curve, and lattice-based math) on ARM chips and FPGAs, catching close to **100% of injected faults** at under **1% added hardware cost**.
- Added hardware-level security (PUF/TRNG) to the post-quantum standards Kyber and Dilithium, speeding up key generation by up to **26%** over the reference design.
- Published 10+ peer-reviewed papers; reviewed 21 manuscripts for IEEE and ACM journals; mentored students in an NSF summer research program.

TD SYNnex

AI Engineer Intern

May 2025-Aug 2025

Clearwater, FL

- Built Jarvis, a multilingual agentic assistant on NVIDIA AI Enterprise, combining document search (RAG) over enterprise files, NVIDIA NIM services, agent memory, and tool calls to the Google Places and Directions APIs, with NeMo Guardrails to constrain topics and reduce hallucinations.

PaarLift

ML Research Engineer

Jan 2018-Apr 2020

Tehran, Iran

- Built ML models (K-nearest neighbors) to position elevators by time of day, cutting passenger wait times **27%** across **100+ commercial buildings**.
- Connected Raspberry Pi boards to elevator controllers over the CAN bus for real-time data into a Linux IoT system, with Apache Airflow ETL pipelines feeding an OLAP store.
- Led a team of 4 software engineers to deliver the product.

SELECTED PROJECTS

1. **job-eval.com** *Founder* | 2K active users GitHub

Trained a 156M-parameter ModernBERT student to score how well a resume matches a job, distilled from a Qwen3.5-9B teacher acting as an LLM judge that generates the training labels. A customized head uses two self-attention pooling methods to weigh each token's importance across four scores - requirement coverage, seniority fit, domain alignment, and ATS keyword overlap. Shipped open-source and as a hosted product with 2K active users.

LLM-as-judge | *knowledge distillation* | *ModernBERT* | *PyTorch*

2. **PII-360** *Open-source* GitHub

Chrome extension that detects and hides personal information in images, text, and PDFs. Runs entirely in the browser with ONNX models, so nothing leaves the user's device.

ONNX | *on-device inference* | *multilingual PII*

3. **Gradient-Based Attacks on RAG Systems** *Open-source research* GitHub

Research showing how small, targeted changes to text embeddings degrade RAG search results and the answers built on them.

Adversarial ML | *retrieval* | *PyTorch*

SELECTED PUBLICATIONS

[Google Scholar](#)

1. **An Interactive Framework for Implementing Privacy-Preserving Federated Learning: Experiments on Large Language Models**

K. Ahmadi et al. | IEEE Security and Privacy Workshops, 2025 | **First author** | **Best Paper Award**

2. **PII Span Localization with a Frozen Autoregressive LLM**

K. Ahmadi et al. | RealmLabs manuscript, 2026 | **First author**

3. **Efficient Algorithm-Level Error Detection for the Number-Theoretic Transform Assessed on FPGAs**

K. Ahmadi, S. Aghapour, M. Mozaffari Kermani, R. Azarderakhsh | ACM Transactions on Embedded Computing Systems, 2025 | **First author**

4. **Efficient Error Detection Schemes for the ECSM Window Method Benchmarked on FPGAs**

K. Ahmadi et al. | IEEE Transactions on VLSI Systems, 2024 | **First author**

TECHNICAL SKILLS

Machine Learning: deep learning, LLMs, model training & fine-tuning (LoRA, DP-SGD), representation learning, evaluation & ablations, PyTorch, Hugging Face, TensorFlow, scikit-learn

LLM Systems: vLLM serving, batching & caching, hidden-state extraction, sparse autoencoders, model probes, RAG, FastAPI, ONNX, Docker, AWS

ML Security & Privacy: PII detection, adversarial ML, differential privacy, federated learning, fault-injection detection

Cryptographic & Hardware Security: side-channel & fault-detection schemes, post-quantum cryptography (Kyber, Dilithium), PUF/TRNG, ARM, FPGA (Xilinx Vivado)

Languages: Python, SQL, C, C++, Verilog, Bash | **Certifications:** AWS Certified Solutions Architect - Associate